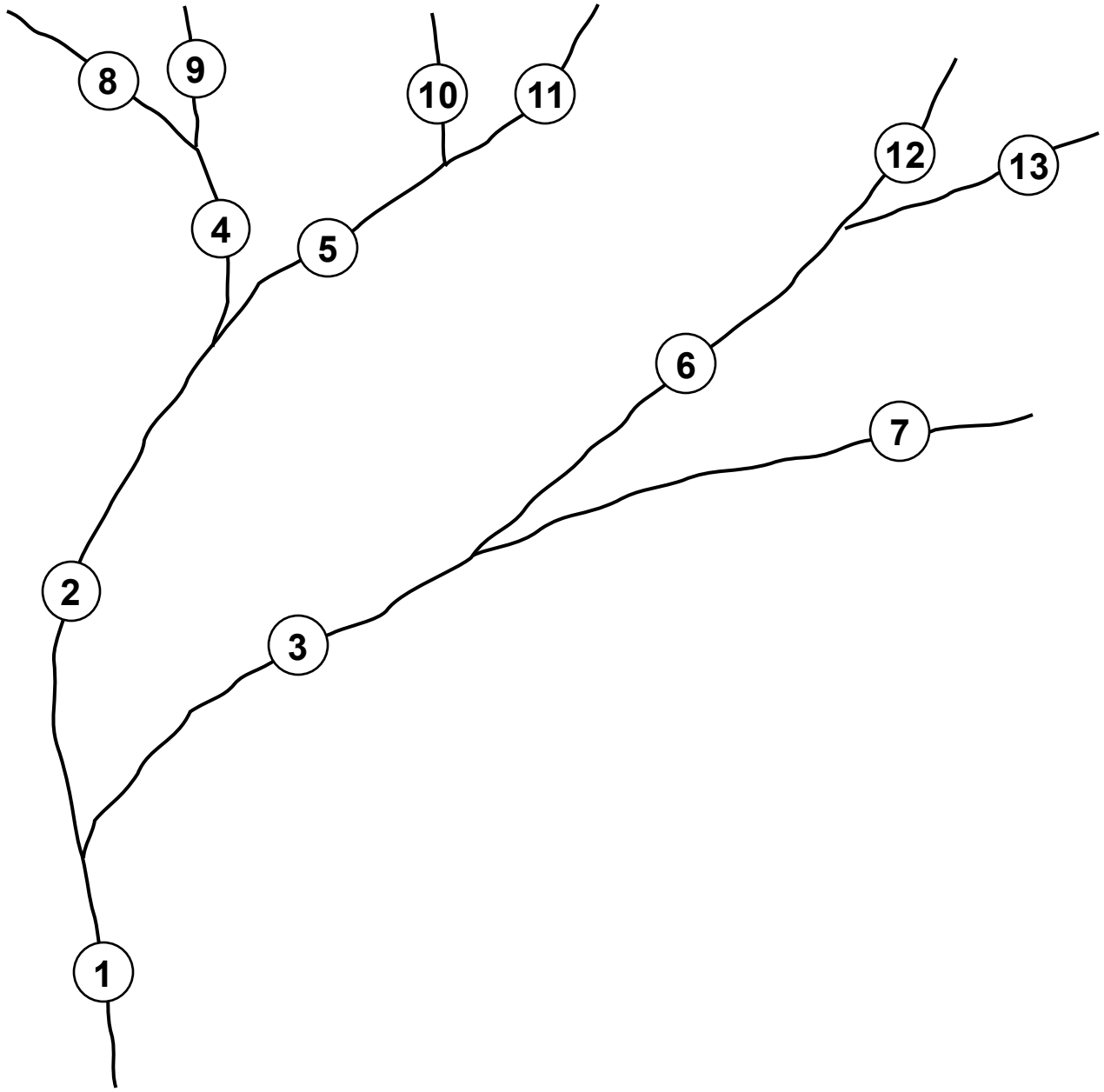
Practical 7: Diamond exploration game

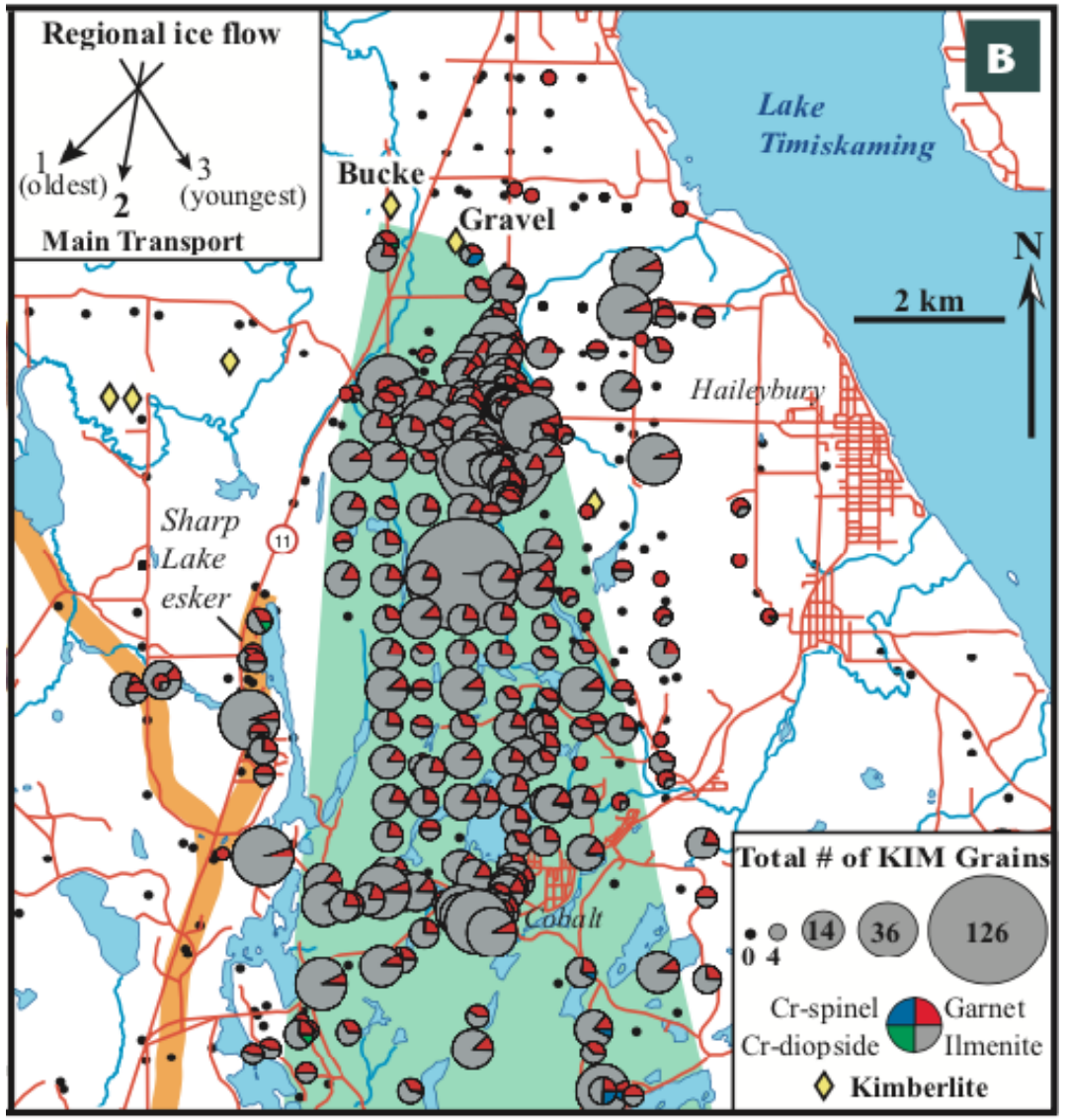
Because diamonds occur in parts per billion in their host rock, geologists hunt for the minerals that commonly form along with the diamonds (called indicator minerals). These are diamond (of course), diopside, garnets, ilmenite, and olivine, and others. Your team of geologists has the task of identifying a number of kimberlite pipes, by analyzing the indicator minerals in placer samples taken along the drainage network, and to assess their commercial value to obtain a production licence. Recall your basic mineral identification from year 1, and you are equally being given an identification sheet to help you.
The map of your field area shows potential sample locations with numbers. As you study the grains from each location record your data in the table provided below. Continue upstream until you find a sample with a higher percent of indicator minerals or a diamond (or run out of money.) All teams start at location 1 and with an exploration budget of $10,000.

Rules:
1. Your team captain needs to contact the demonstrators or myself to obtain a new sample. Each sample costs $500.
2. Your group can only have one sample at a time. For each sample you must identify the materials, record your data in the table, and check your answers with the demonstrators or myself. You can only proceed if all materials are identified correctly.
3. Before you can receive your next sample you must give us a justification for selecting the next location and balance your exploration budget.
4. Continue upstream towards the diamond source(s), unless your data suggest you should move to another stream. There are 3 pipes in the area. You can try to identify all if you have time and money left.
5. If you have identified a pipe, you can commission a drilling campaign (an absolute bargain at $500 per pipe) to assess the value (in $ per tonne, see lecture). This is optional.
5. After the time is up, each team can bid for one or more pipes to obtain a production licence.
6. After licenses are awarded, we will calculate your team's profit (Income from license - money spent during exploration + leftover exploration budget). Winner is the team with the highest profit.


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Exploration
Indicator mineral sampling





- Analysis of heavy mineral placers directly associated with kimberlite

Kjarsgaard 2019

Sample	Diamond	Diopside	Garnet	Ilmenite	K Feldspar	Olivine	Quartz	Garnet schist	Amphibolite schist	Sediment. rock	Halite	Muscovite	Biotite
1													
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													